

KISUMU, Kenya

WMO: 637080

Lat: 0.086S

Lon: 34.729E

Elev: 1209

StdP: 87.62

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	16.4	7.7	7.5	27.6	9.6	8.6	25.8	10.0	24.4	8.6	25.0	1.0	60	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.8	33.2	18.9	32.1	19.3	31.1	19.7	22.7	28.0	22.2	27.7	21.9	27.5	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	18.3	25.3	20.6	17.7	24.7	20.2	17.2	24.2	73.9	28.0	71.9	28.1	70.4	27.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.6	7.6	12.1	35.4	2.1	1.2	10.6	36.3	9.4	36.9	8.2	37.6	6.7	38.5		
(5)			10.6	25.1	1.2	1.7	9.8	26.3	9.0	27.3	8.4	28.3	7.5	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	24.0	24.7	24.6	23.8	23.2	22.8	22.7	23.1	23.5	24.0	23.6	23.7
	DBStd	1.27	1.25	1.35	1.47	1.05	0.80	0.86	1.04	1.00	1.08	1.10	1.05	1.18
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	4982	436	412	453	413	411	385	393	404	406	434	409	426
	CDD18.3	1940	177	178	195	163	152	135	135	146	156	176	159	167
	CDH23.3	13564	1476	1631	1592	1017	847	785	812	905	1055	1194	1009	1243
CDH26.7	4051	531	681	607	255	153	132	156	213	317	367	259	382	
Wind	WSAvg	2.6	3.1	3.2	3.1	2.5	2.2	2.2	2.3	2.5	2.7	2.6	2.4	2.7
Precipitation	PrecAvg	1334	60	97	143	212	172	74	61	110	91	80	137	103
	PrecMax	1635	244	268	286	358	320	140	140	226	167	181	464	277
	PrecMin	950	5	11	1	62	77	26	13	27	7	17	41	23
	PrecStd	200	56	70	78	78	66	29	35	58	47	46	99	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.9	34.9	31.9	30.0	29.7	30.1	31.0	32.7	32.2	31.5	33.2
		MCWB	18.1	18.0	18.3	20.2	20.7	19.6	19.0	18.8	19.2	19.5	19.5	18.7
	2%	DB	32.1	33.2	32.8	30.2	29.0	28.9	29.1	29.9	30.9	31.1	30.1	31.2
		MCWB	18.9	18.6	19.6	20.9	20.7	20.0	19.1	19.4	19.4	19.8	20.1	19.2
	5%	DB	30.9	32.0	31.2	29.1	28.2	28.1	28.2	28.9	29.8	30.0	29.2	30.0
		MCWB	19.3	19.3	20.3	21.0	20.8	20.1	19.4	19.6	19.8	20.0	20.3	19.6
	10%	DB	29.8	30.8	30.0	28.1	27.5	27.2	27.5	28.0	28.6	28.9	28.2	28.9
		MCWB	19.9	19.7	20.5	21.0	20.8	20.1	19.5	19.8	20.0	20.1	20.4	20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.8	23.0	23.1	22.6	22.0	21.5	21.8	22.3	22.4	23.0	22.6
		MCDB	28.1	29.5	28.9	28.2	27.2	27.3	27.2	27.4	28.1	27.6	27.4	27.3
	2%	WB	21.9	22.0	22.3	22.4	22.0	21.3	20.9	21.0	21.6	21.7	21.9	21.8
		MCDB	27.7	28.5	28.3	27.5	27.0	26.7	26.7	27.3	27.6	27.5	27.2	27.0
	5%	WB	21.4	21.4	21.8	21.9	21.5	20.9	20.4	20.5	21.0	21.1	21.3	21.3
		MCDB	27.2	27.8	27.7	27.1	26.5	26.4	26.1	26.7	27.3	27.2	26.7	26.5
	10%	WB	20.9	20.9	21.3	21.4	21.1	20.5	20.0	20.0	20.4	20.6	20.8	20.9
		MCDB	26.7	27.3	27.1	26.4	26.1	25.8	25.7	26.2	26.6	26.5	26.2	26.0
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.8	11.7	10.0	9.7	10.2	10.5	11.0	11.6	11.3	10.6	11.0
		MCDBR	14.2	15.2	13.6	11.3	10.5	11.1	11.7	12.2	13.3	12.7	11.7	13.0
		MCWBR	6.1	6.2	5.5	4.4	4.2	4.5	4.5	4.6	5.1	4.6	4.5	5.4
	5% WB	MCDBR	11.7	12.2	11.5	10.1	9.6	9.9	10.4	11.0	11.4	11.0	10.4	10.5
		MCWBR	5.3	5.4	5.0	4.4	4.3	4.4	4.5	4.7	5.1	4.6	4.5	4.8
Clear-Sky Solar Irradiance	taub	0.385	0.409	0.410	0.388	0.388	0.447	0.461	0.438	0.405	0.392	0.379	0.382	
	taud	2.428	2.353	2.374	2.466	2.462	2.261	2.202	2.274	2.372	2.412	2.469	2.448	
	Ebn at Noon	944	924	914	914	891	824	817	857	905	927	943	942	
	Edh at Noon	120	132	128	114	110	132	142	136	127	122	115	117	
All-Sky Solar Radiation	RadAvg	6.22	6.53	6.43	5.92	5.76	5.42	5.33	5.61	6.15	6.02	5.72	5.92	
	RadStd	0.41	0.32	0.42	0.36	0.20	0.28	0.27	0.25	0.23	0.24	0.32	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page